Basic Text Generation

```
from transformers import AutoModelForCausalLM, AutoTokenizer
import torch

def generate_response(model_name, prompt):
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    model = AutoModelForCausalLM.from_pretrained(model_name)
    inputs = tokenizer(prompt, return_tensors="pt")
    outputs = model.generate(inputs['input_ids'], max_length=100)
    return tokenizer.decode(outputs[0], skip_special_tokens=True)

prompt_1 = "List the top three uses of apple."
print("GPT-Neo 1.3B response:", generate_response("EleutherAI/gpt-neo-1.3B", prompt_1))
```

Modified Prompts Comparison

```
prompt_fruit = "List the top three uses of apple fruit."
prompt_drink = "List the top three uses of apple drink."

print("=== Apple Fruit Prompt ===")
response_fruit = generate_response("EleutherAI/gpt-neo-1.3B", prompt_fruit)
print(response_fruit)

print("\n=== Apple Drink Prompt ===")
```

```
response_drink = generate_response("EleutherAI/gpt-neo-1.3B", prompt_drink)
print(response_drink)
```

Enhanced Version with Better Parameters

```
from transformers import AutoModelForCausalLM, AutoTokenizer
import torch

def generate_response_enhanced(model_name, prompt, max_length=100, temperature=0.7, do_sample=True):
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    model = AutoModelForCausalLM.from_pretrained(model_name)

    # Add padding token if it doesn't exist
    if tokenizer.pad_token is None:
        tokenizer.pad_token = tokenizer.eos_token

    inputs = tokenizer(prompt, return_tensors="pt", padding=True, truncation=True)

    # Generate with more control parameters
    outputs = model.generate(
        inputs['input_ids'],
        max_length=max_length,
        temperature=temperature,
        do_sample=do_sample,
        pad_token_id=tokenizer.eos_token_id,
```

```
        attention_mask=inputs['attention_mask']
    )

    return tokenizer.decode(outputs[0], skip_special_tokens=True)

# Test with different prompts
prompts = [
    "List the top three uses of apple.",
    "List the top three uses of apple fruit.",
    "List the top three uses of apple drink."
]

for i, prompt in enumerate(prompts, 1):
    print(f"\n=== Prompt {i}: {prompt} ===")
    response = generate_response_enhanced("EleutherAI/gpt-neo-1.3B", prompt)
    print(response)
```